Turja Kundu

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Research Focus

High-performance computing; scalable graph learning; temporal/dynamic graph learning.

Reference Dr. Sanjukta Bhowmick (Advisor) `Sanjukta.Bhowmick@unt.edu`; Dr. Yuede Ji `yuede.ji@uta.edu`

Education

2021–Present **Ph.D. Candidate, Computer Science and Engineering**, *University of North Texas*, Initiated at SIUC

2007–2012 **B.Sc. in Computer Science and Engineering**, *Bangladesh Univ. of Engineering and Technology (BUET)*

Internship

Summer 2026 **Computational Science Intern**, *Fermi National Accelerator Laboratory (Fermilab)*, Batavia, IL
Building a scalable CVN neutrino-event inference pipeline by integrating LArSoft/larrecodnn with NuSONIC/NVIDIA Triton and benchmarking latency/throughput.

Research Experience

ATLAS **ATLAS: Adaptive Topology-based Learning at Scale for Homophilic and Heterophilic Graphs** (under review at KDD). Proposed a propagation-free, topology-augmented MLP using adaptive multi-resolution community features for scalable i.i.d. minibatch training and adjacency-free inference.

HIOS **HIOS: Hierarchical Inter-Operator Scheduler for Real-Time Inference of DAG-Structured Deep Learning Models on Multiple GPUs** (IEEE Cluster 2023). Developed dependency-aware inter/intra-GPU scheduling to reduce inference latency.

Centrality **Centrality Resilience in Complex Networks** (ICCS 2025). Studied preservation of closeness/betweenness rankings under targeted edge attacks against top-k high-centrality vertices.

Publications

Under Review **T. Kundu**, S. Bhowmick. *ATLAS: Adaptive Topology-based Learning at Scale for Homophilic and Heterophilic Graphs.* `https://arxiv.org/html/2512.14908v5`

Conf. **T. Kundu**, T. Shu. *HIOS: Hierarchical Inter-Operator Scheduler for Real-Time Inference of DAG-Structured Deep Learning Models on Multiple GPUs.* IEEE Cluster 2023.

F. A. Irany, **T. Kundu**, S. Sarakar, A. Mukherjee, S. Bhowmick. *Centrality Resilience in Complex Networks.* ICCS 2025, Short Paper.

Workshop Y. Zhang, D. Pandey, D. Wu, **T. Kundu**, R. Li, T. Shu. *Accuracy-Constrained Efficiency Optimization and GPU Profiling of CNN Inference.* SC Workshops 2023.

Poster **T. Kundu**, S. Bhowmick. *Effect of Graph Structure vs. Feature Quality for GCN.* SIAM 2025.

Teaching and Mentoring Experience

TA Southern Illinois University Carbondale; University of North Texas, 2021–2024. Courses: Linux Programming; Computer Organization; Distributed Programming; Advanced OOP; Computer Architecture; Operating Systems; Graph Theory; Algorithms; Discovering Computer Science.

Coursework and Technical Skills

Coursework Deep Learning; Distributed Systems; Numerical Computation; Graph Theory; Software–Hardware Codesign for DNN; Computer Architecture; Real-Time Operating Systems; Computational Statistics.

Skills C, C++, CUDA, MPI, OpenMP, Python, Java, Fortran, R, Bash; PyTorch, cuDNN, NCCL, CUPTI, ROS, NetworkX, scikit-learn, NumPy, Pandas; Nsight Systems/Compute, `perf`, Wireshark.

Awards

NSF Travel Grant, IEEE Cluster 2023; Dean's List Award, BUET, 2010.

Industry Experience

2017–2021 **Assistant Director (IT)**, ASA International Management Services Ltd, Dhaka – Developed web-based microfinance banking systems and led large-scale data migration.

2015–2017 **Software Engineer**, Relisource Technology Ltd, Dhaka – Built distributed service-oriented systems using Windows Services.

2012–2015 **Software Engineer**, Mir Technology Ltd, Dhaka – Developed SIP relay/proxy servers using pthreads/BSD sockets and Android SIP clients.